

# Deck Number Check

Verification of figures proposed for a client deck,  
with an internally consistent replacement set

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## Client

Iwatani Corporation — Metals Department → Stainless Steel Division

## Subject

Bellows tube (precision-slit material) market entry,  
leveraging the Jindal Stainless relationship

## Status

Working document — updated at monthly internal meetings

## Prepared

July 2026

**Evidence standard.** Every factual claim in this document carries an inline source link and a verification grade: **[P]** primary (company, association or government publication), **[C]** credible press, **[S]** commercial market-research vendor (indicative only), **[U]** unverified directory listing. Commercial estimates for the bellows market disagree with one another by up to a factor of 19; figures marked **[S]** must not be quoted externally without stating the range. One proposed figure does not reconcile and should not be used as a point estimate.

# 13 — Deck Number Check

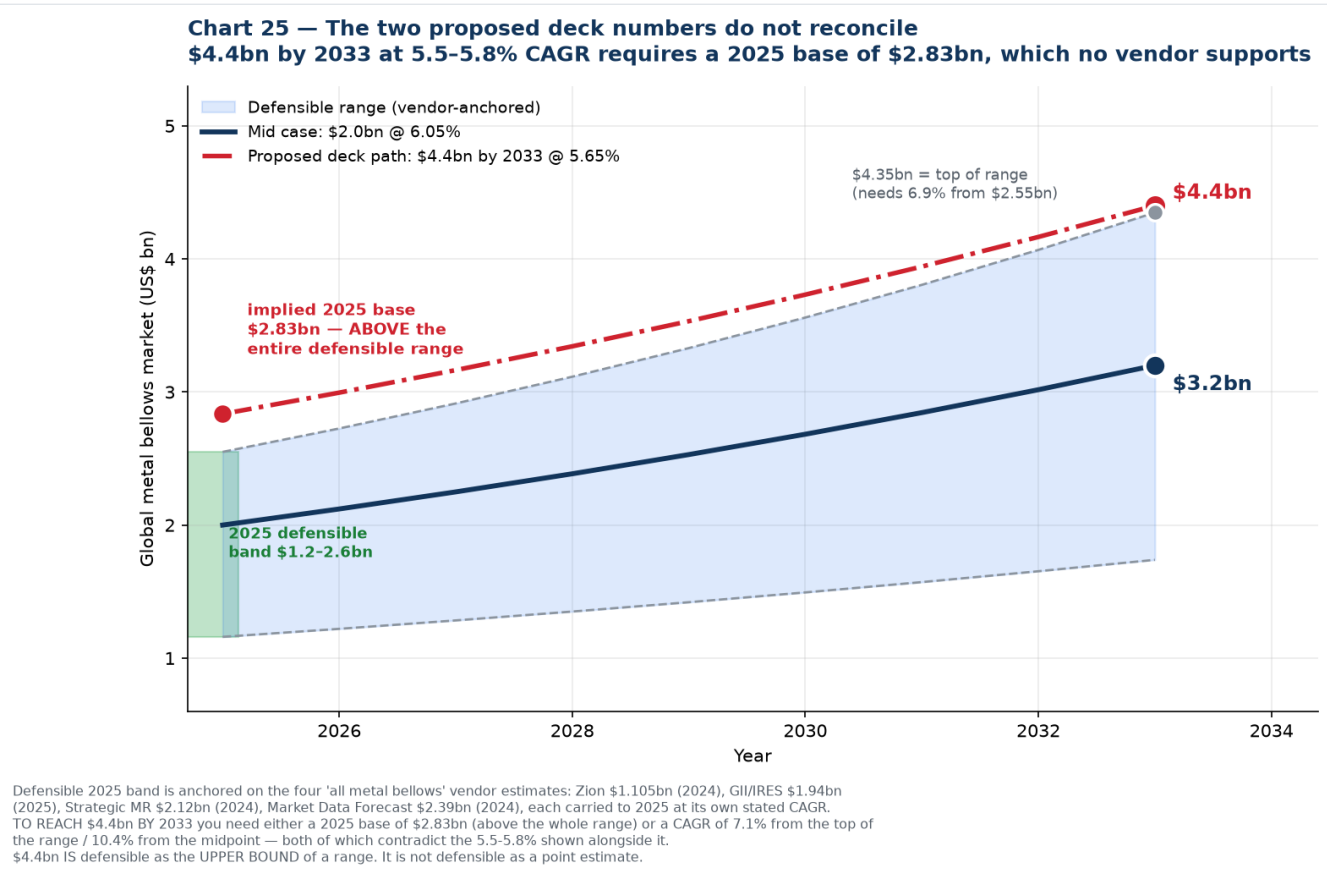
Verification of four figures proposed for a client deck.

## Verdict at a glance

| Proposed figure                           | Verdict                                                                                |
|-------------------------------------------|----------------------------------------------------------------------------------------|
| Global metal bellows, 2025 — (left blank) | Use <b>US\$1.2-2.6 bn</b> , midpoint ~ <b>US\$2.0 bn</b>                               |
| Forecast 2033: ~ <b>US\$4.4 bn</b>        | <b>Do not use as a point estimate.</b> Defensible only as the top of a range           |
| CAGR 2025-33: ~ <b>5.5-5.8%</b>           | <b>Confirmed.</b> Well supported — this is the modal cluster                           |
| India global share: ~ <b>2.7%</b>         | <b>Confirmed as an estimate.</b> Must be labelled modelled; no published figure exists |

The core problem: the two numbers you have contradict each other.

## 1. Why US\$4.4 bn by 2033 fails



If 2033 is US\$4.4 bn and the CAGR is 5.5-5.8%, then working backwards the **2025 base must be US\$2.80-2.87 bn.**

That is above every vendor estimate for the global metal bellows market:

| Vendor                      | Base-year value     | Carried to 2025 at its own CAGR |
|-----------------------------|---------------------|---------------------------------|
| Zion Market Research        | US\$1.105 bn (2024) | US\$1.16 bn                     |
| GII / IRES                  | US\$1.94 bn (2025)  | US\$1.94 bn                     |
| Strategic Market Research   | US\$2.12 bn (2024)  | US\$2.27 bn                     |
| Market Data Forecast        | US\$2.39 bn (2024)  | US\$2.53 bn                     |
| <b>Defensible 2025 band</b> |                     | <b>US\$1.16 - 2.55 bn</b>       |

US\$2.83 bn sits **11% above the top of the range and 42% above the midpoint.**

### Put the other way: what CAGR would you need to reach US\$4.4 bn?

| Starting from 2025 base of... | Required CAGR to 2033 |
|-------------------------------|-----------------------|
| US\$1.16 bn (low)             | <b>18.1%</b>          |
| US\$2.00 bn (mid)             | <b>10.4%</b>          |
| US\$2.39 bn                   | <b>7.9%</b>           |
| US\$2.55 bn (high)            | <b>7.1%</b>           |

Every one of those exceeds the 5.5–5.8% written beside it. **You cannot hold both numbers at once.**

### What US\$4.4 bn is defensible as

The top of the range. From the highest defensible 2025 base (US\$2.55 bn) at the highest vendor CAGR (6.90%), 2033 comes out at **US\$4.35 bn ≈ US\$4.4 bn**. So this phrasing works:

"The global metal bellows market is estimated at US\$1.2–2.6 bn in 2025, growing at 5–7% a year to somewhere between **US\$1.7 bn and US\$4.4 bn by 2033.**"

What does not work is "the market will reach US\$4.4 bn by 2033," because that presents the most optimistic corner of a wide band as a central case, and it can be dismantled by anyone who divides one number by the other.

## 2. CAGR 5.5–5.8% — confirmed

This is the strongest of the four figures. It is not just inside the consensus band, it is the **modal cluster** of published estimates:

| Vendor                      | CAGR         |
|-----------------------------|--------------|
| Zion Market Research        | 5.19%        |
| Market Research Future      | 5.60%        |
| <b>Market Data Forecast</b> | <b>5.74%</b> |
| <b>GII / IRES</b>           | <b>5.79%</b> |
| Expert Market Research      | 5.90%        |
| Dataintelo (welded)         | 6.20%        |
| Dataintelo (hydroformed)    | 6.80%        |
| Strategic MR / SkyQuest     | 6.90%        |

Five of eight land between 5.19% and 5.90%. **5.5–5.8% is well chosen.** If anything it is slightly conservative; ~6% is the true midpoint. Either is defensible.

### 3. India share ~2.7% — confirmed, with a labelling requirement

Correct as an estimate, and it is our figure. Two conditions on using it:

- 1. **Label it as an estimate.** No research vendor publishes an India bellows market share. Phrase it as "estimated at approximately 2–3%" rather than stating 2.7% as fact.
- 2. **It only reconciles with a ~US\$2.0 bn global base.** Cross-check: 2.7% of US\$2.0 bn = **US\$54 m**, which sits inside our independently derived India estimate of US\$50–70 m. But 2.7% of US\$2.83 bn (the base implied by your US\$4.4 bn figure) = **US\$76 m**, which falls outside that range. The inflated global number breaks the India number downstream too.

This is worth noting because it shows the error compounds: one over-stated figure makes two other slides inconsistent.

### 4. If you need a single citable source with a name on it

For a deck, a named source often beats a modelled range. The highest **sourced** 2033 figure for the all-metal-bellows scope is:

**Market Data Forecast: US\$2.39 bn (2024) → US\$3.95 bn (2033), implied CAGR 5.74%**

That gives you a 2033 number close to US\$4 bn, a CAGR inside your 5.5–5.8% target, and an attributable source. If the deck needs one line with a citation, use this and name the firm — adding that other estimates range from US\$1.1 bn to US\$2.4 bn for the base year because scope definitions differ.

### 5. Deck-ready numbers

| DECK-READY NUMBERS — internally consistent, each traceable to a source                                                                                                                                                                              |                  |                               |                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------|--------------------------------------------------|
| Metal bellows market · use these exact figures and phrasings                                                                                                                                                                                        |                  |                               |                                                  |
| Global market size, 2025                                                                                                                                                                                                                            | US\$1.2 - 2.6 bn | midpoint ~US\$2.0 bn          | Range of 4 vendors; scope definitions differ     |
| Global forecast, 2033                                                                                                                                                                                                                               | ~US\$3.2 bn      | range US\$1.7 - 4.4 bn        | Modelled from the 2025 band at consensus CAGR    |
| CAGR, 2025 - 2033                                                                                                                                                                                                                                   | ~6%              | range 5.2 - 6.9%              | Every vendor falls inside this band              |
| Highest SOURCED 2033 figure                                                                                                                                                                                                                         | US\$3.95 bn      | Market Data Forecast          | US\$2.39bn (2024) at 5.74% — cite by name        |
| India market size, 2025                                                                                                                                                                                                                             | US\$50 - 70 m    | ≈ ₹435 - 610 cr               | Bottom-up from MCA filings + 3 vendors           |
| India share of global                                                                                                                                                                                                                               | ~2 - 3%          | point estimate 2.7%           | MODELLED — no published figure exists            |
| India growth, observed                                                                                                                                                                                                                              | +6.5%            | FY24 → FY25, revenue-weighted | Measured from company filings — strongest number |
| DO NOT PUT "US\$4.4bn by 2033" ON A SLIDE AS A POINT ESTIMATE. It implies a 2025 base of US\$2.83bn, above every vendor estimate, and contradicts the 5.5-5.8% CAGR beside it. Use "~US\$3.2bn" as the figure, or "up to US\$4.4bn" as a range top. |                  |                               |                                                  |

| Metric                      | Use this                | Range                         | Basis                                         |
|-----------------------------|-------------------------|-------------------------------|-----------------------------------------------|
| Global market size, 2025    | <b>US\$1.2 - 2.6 bn</b> | midpoint ~US\$2.0 bn          | Four vendors; scope definitions differ        |
| Global forecast, 2033       | <b>~US\$3.2 bn</b>      | US\$1.7 - 4.4 bn              | Modelled from the 2025 band at consensus CAGR |
| CAGR, 2025-33               | <b>~6%</b>              | 5.2 - 6.9%                    | Every vendor falls inside this band           |
| Highest sourced 2033 figure | <b>US\$3.95 bn</b>      | —                             | Market Data Forecast, cite by name            |
| India market size, 2025     | <b>US\$50 - 70 m</b>    | ≈ ₹435 - 610 cr               | Bottom-up from MCA filings + three vendors    |
| India share of global       | <b>~2 - 3%</b>          | point estimate 2.7%           | <b>Modelled</b> — no published figure exists  |
| India growth, observed      | <b>+6.5%</b>            | FY24 → FY25, revenue-weighted | Measured from company filings                 |

### Suggested slide wording

**Global metal bellows market** US\$1.2-2.6 bn (2025) · growing ~6% p.a. · ~US\$3.2 bn by 2033  
Estimates vary by scope; range reflects four independent vendors

**India** US\$50-70 m (2025) · ~2-3% of global · +6.5% observed FY24→FY25 India share is our estimate; no published figure exists

Both lines survive scrutiny, and the caveats are short enough to sit in a footnote.

## 6. Two other figures to keep off the deck

Carried forward from earlier corrections in this pack, in case they are still circulating:

| Figure                                                    | Status                                                                                                                     |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| "India holds 5-10% of the global expansion joints market" | <b>Withdrawn.</b> Implies a global market of US\$0.7-1.4 bn against four vendors at US\$3.7-7.2 bn. See <a href="#">11</a> |
| "India bellows CAGR ~6.9% (6Wresearch)"                   | <b>Withdrawn.</b> 6Wresearch publishes no market CAGR. See <a href="#">10</a> §10                                          |
| "North American semiconductor bellows market US\$1.2 bn"  | <b>Never use.</b> Exceeds half the global all-industry market                                                              |
| "48% semiconductor + 56% aerospace of US demand"          | <b>Never use.</b> Sums to 104%                                                                                             |

## 7. The check to run on any future number

Before a figure goes on a slide, do the one-line reconciliation: **divide the forecast by the base and see whether the implied CAGR matches the CAGR you are quoting.** If they disagree, one of the three numbers is wrong.

That single division would have caught the US\$4.4 bn figure, the 5-10% expansion joints claim and the 104% end-use split. It takes ten seconds and it is the difference between a number that holds up in a partner meeting and one that does not.

## Chart index

| Chart | File                                     | Purpose                                      |
|-------|------------------------------------------|----------------------------------------------|
| 25    | <a href="#">25-deck-number-check.png</a> | Why US\$4.4 bn and 5.5–5.8% cannot both hold |
| 26    | <a href="#">26-deck-card.png</a>         | Deck-ready figures, internally consistent    |

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